Offline Rowing IMU Dashboard - User Guide

Purpose: prepare a SEAT and BOAT IMU recording so stroke detection is plausible and coaches or athletes can understand the most important setup steps. All dashboard speed values are velocity proxies estimated from acceleration. They are useful for timing and comparison, but they are not calibrated physical speed in m/s.

Filter note: the current system does not use a Kalman filter. The current pipeline removes a baseline, smooths acceleration with a moving average, integrates acceleration into a velocity proxy, corrects simple linear drift, and then detects stroke starts from the local velocity curve.

1. Record and open the report

1. Turn on both IMUs and record the session. The SD cards do not need to be connected at the same time later. Copy the CSV files into one folder.

2. Run the offline analysis with that folder as the input. The tool creates a self-contained HTML file. This file can be opened without Python and can be sent to another person.

3. First check whether SEAT and BOAT were detected correctly. If the files were swapped, use swap SEAT/BOAT data in the Sensor alignment section.

2. Sensor alignment

Setting	When to use it	Effect
Forward axis	Use this when the IMU is mounted differently and another axis points along the rowing direction.	Selects the axis used as forward movement. The current default is Y lateral as forward.
invert forward sign	Use this if the curves look logically mirrored.	Flips the sign for SEAT and BOAT together.
invert SEAT only	Use this if only the seat sensor was mounted in the opposite direction.	Flips only the SEAT contribution. BOAT stays unchanged.
swap SEAT/BOAT data	Use this if the CSV files were detected or selected in the wrong order.	Swaps the two data sources inside the dashboard.

3. Stroke detection tuning in the first minute

The dashboard defines one stroke window from one detected drive start to the next detected drive start. In other words: stroke time is measured drive start to drive start. For a new mounting setup or athlete, manually count the real rowing strokes in the first minute and adjust the dashboard settings until the displayed count matches your manual count as closely as possible.

Step	Action	Meaning
1	Zoom the Time graph to the first 60 seconds or set the zoom fields to 0 and 60.	You compare dashboard detection and manual counting in the same time window.
2	Manually count the strokes in that minute.	This is the reference count for tuning.
3	Check Detected strokes in the dashboard.	If the dashboard counts too many strokes, it probably counts bumps as extra strokes. If it counts too few, detection is too strict for fast strokes.
4	Adjust Stroke detection: minimum gap ms.	Higher reduces double-counting. Lower allows faster strokes to be detected.
5	Adjust Stroke detection: smoothing samples.	Higher makes curves calmer. Lower is more responsive, but more sensitive to vibration.
6	Repeat until manual stroke count and dashboard stroke count are close.	Use the same settings for this recording afterwards.

Suggested starting point: minimum gap 800 ms and smoothing 10 samples. If fast strokes are missed, first lower the minimum gap. If too many false strokes are detected, first increase minimum gap or smoothing.

4. Athlete segments and export

If multiple athletes row one after another, keep everything in one long recording. In the coach dashboard, mark a time range in the graph or type it into the zoom fields, then click Save visible time range as athlete. Each athlete receives a separate segment row.

Before exporting HTML, the coach can write a short message for the athlete. The exported athlete report shows the same main graphs as the coach report, but time is local to that athlete. A segment from 60 to 110 seconds is shown as 0 to 50 seconds in the athlete report.